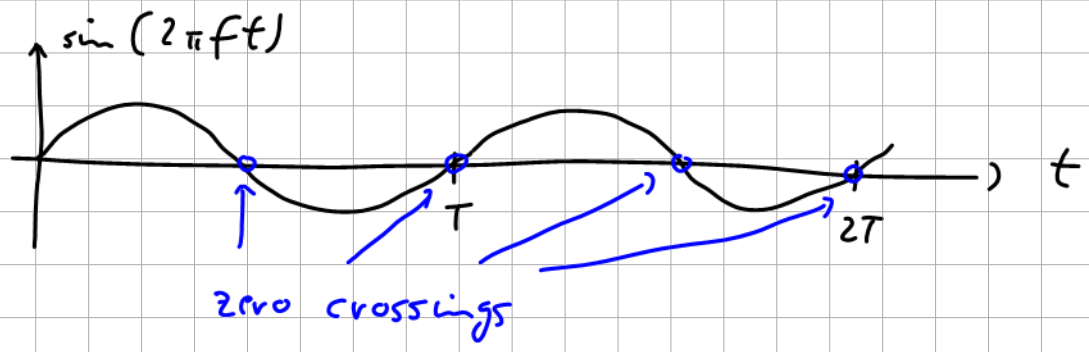
1)



Obviously one period T of a sinus has 2 zero crossings.

$\Rightarrow$ $f = 1234$ Hz $\Rightarrow$ 2468 zero crossing per second

2 zero crossings for one period length

one period length is $\tau = \frac{1}{f}$ in seconds or

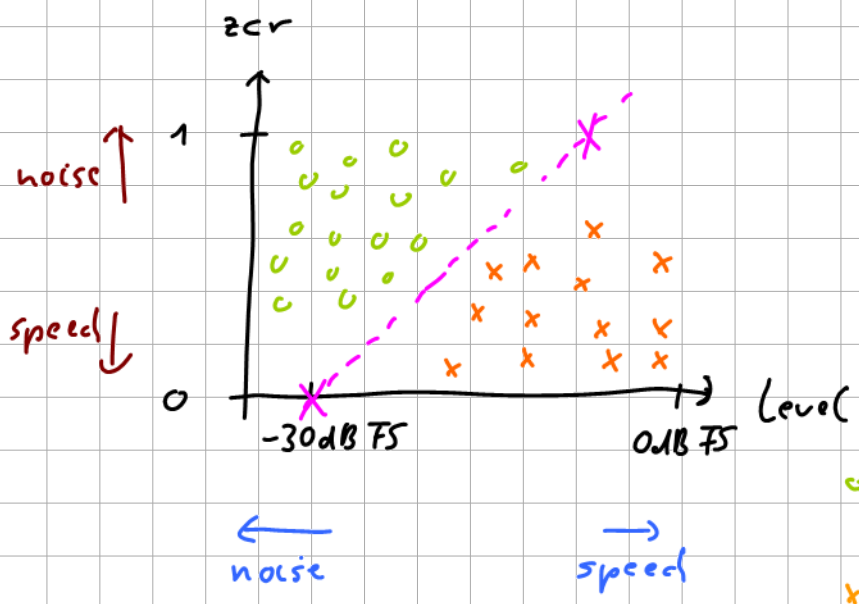
$\frac{r}{f}$ in samples

$\Rightarrow$ $zcr = \frac{2f}{r}$

e.g. for a sampling rate of $r = 16000$

$$zcr = \frac{2 \cdot 1234}{16000}$$

2) Each feature form one dimension of the so called „feature space":



The louder a signal, the higher the probability, that speech is active.

The lower the zero crossing rate, the higher the probability that speech is active.

o Assumed features for noisy frames

x Assumed features for speech frames

In the shown feature space, noise- and speech-frames can be separated by the dashed line.

The proposed linear separator can be described by two points marked with X:

$$(L = -30 \text{ dB FS} \mid zcr = 0)$$

$$(L = -7 \text{ dB FS} \mid zcr = 1)$$

From this the linear separator can be described by the following equation:

$$zcr = m \cdot L + n$$

$$m = \frac{1 - 0}{-7 - (-30)} = \frac{1}{23}$$

$$0 = \frac{1}{23} \cdot (-30) + n \Rightarrow n = \frac{30}{23}$$

$$\Rightarrow zcr = \frac{1}{23} \cdot L + \frac{30}{23}$$

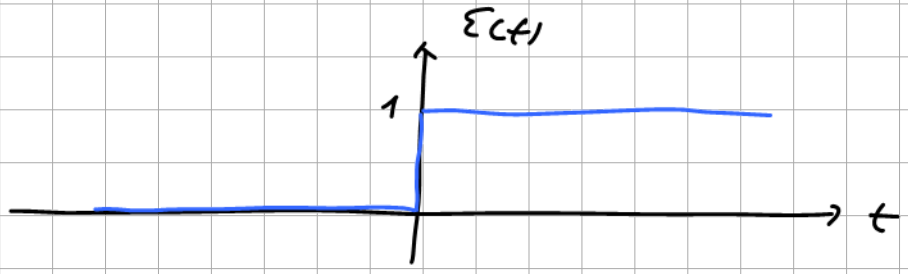
A noisy frame lies above this linear separator:

$\mathcal{D}_s$ Noise : $zcr \geq \frac{1}{23} L + \frac{30}{23}$

A speed frame lies below this linear separator:

$\mathcal{D}_s$ Speed : $zcr < \frac{1}{23} L + \frac{30}{23}$

3)



linearity: $\varepsilon(a x_1 + b x_2) = a \cdot \varepsilon(x_1) + b \cdot \varepsilon(x_2)$

In order to prove non-linearity, one counter example for the above given equation is sufficient:

$a = 1 \qquad x_1 = 2$
$b = -1 \qquad x_2 = 1$

$$\varepsilon(1 \cdot 2 + (-1) \cdot 1) = 1 \cdot \varepsilon(2) - 1 \cdot \varepsilon(1)$$

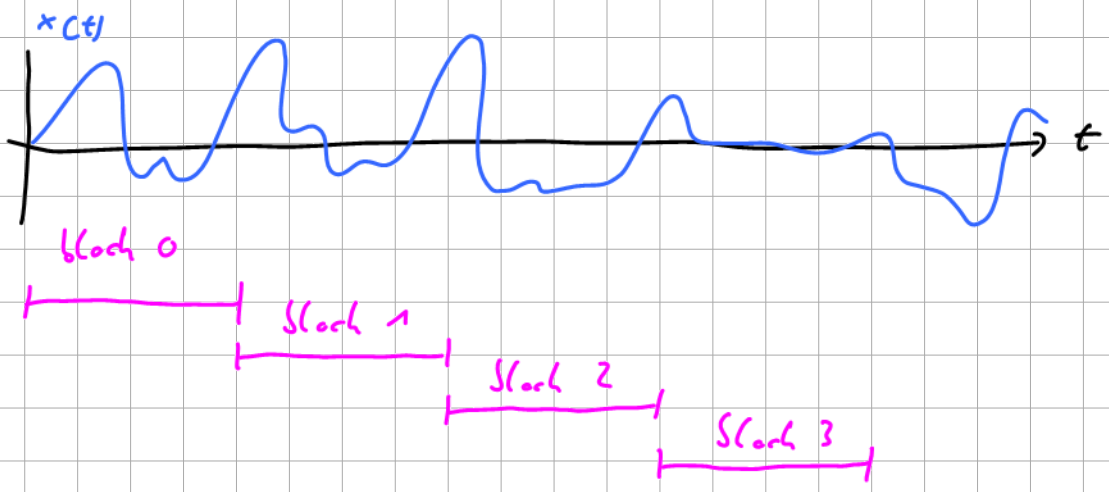
$$\varepsilon(1) = 1 - 1$$

$$1 = 0 \quad \lightning$$

non-linearity is proven!

4) numpy.arange(5) $\hat{=}$ [0 1 2 3 4]

Assuming a signal $x(t)$ and blocks of a given length in milliseconds:



The features are assumed to be valid for the whole block.

In order to align the signal $x(n)$ and the features properly, the feature is assumed to be located in the middle of the block.

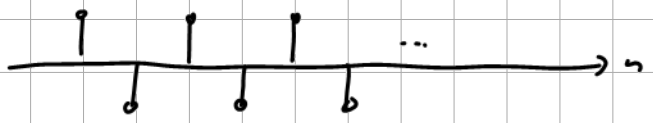
The middle of the block n is:

$(n + 0{,}5) \cdot T$ with T corresponding to the blocksize in seconds.

5) smallest possible ZCR: no zero crossings in the observed block:

$$\underline{\underline{ZCR = 0}}$$

highest possible ZCR: each sample changes the sign:



$$\Rightarrow \quad \underline{\underline{ZCR = 1}}$$